

AI Support Ticket Triage & Human-in-the-Loop Workflow — Overview

Purpose

This workflow automates support ticket classification, routing, email responses, and audit logging using AI predictions and conditional paths. High-risk or sensitive tickets are reviewed by a human before sending replies.

Trigger

Updated Spreadsheet Row

(When a ticket is updated with AI predictions or approval status)

AI Prediction Step

The AI model analyzes the ticket and predicts:

- Issue_Category
- Priority
- Sentiment
- Confidence
- Customer_Risk
- Recommended_Action
- AI_Reply
- Reason

These predictions determine which path the ticket follows.

Paths Overview

1. Routine Support Ticket Path

Condition:

- Issue_Category ≠ Billing
- Customer_Risk = Low or Medium
- Sentiment ≠ Negative
- No human approval required

Actions:

- Update ticket with final resolution
- Send automated support email
- Create audit log entry

2. Billing Escalation Path**Condition:**

- Issue_Category = Billing

Actions:

- Update ticket with billing escalation resolution
- Send billing escalation email
- Create audit log entry

3. High-Risk / Cancellation / Negative Sentiment Path**Condition:**

- Customer_Risk = High
- OR Issue_Category = Cancellation
- OR Sentiment = Negative

Actions:

- Request Human Approval
- **If Approved:**
 - Update ticket
 - Send approved email
 - Create audit log entry
- **If Rejected:**
 - Update ticket
 - No email sent
 - Create audit log entry
 - Assign to Manual Review

End Result

The workflow ensures:

- Routine tickets are handled automatically
- Billing issues are escalated properly

- High-risk tickets get human review
- Every decision is logged for compliance

Future Improvements & Enhancements

Although the current workflow successfully automates ticket triage, routing, human approval, and audit logging, several enhancements could further strengthen the system and make it more scalable, intelligent, and enterprise-ready.

1. Add More Specialized Paths

Right now, the workflow has three main paths: Routine, Billing Escalation, and High-Risk/Cancellation.

You could expand this by adding more specialized paths such as:

Technical Support Path

- For issues related to bugs, errors, or troubleshooting.

Account Management Path

- For subscription upgrades, profile changes, or plan inquiries.

These additional paths would allow more granular routing and more accurate automated responses.

2. Add Audit Logs to Routine & Billing Paths

Currently, audit logs are added primarily for High-Risk/Cancellation tickets.

To improve compliance and traceability:

- Add audit logging to **Routine Path**
- Add audit logging to **Billing Escalation Path**

This ensures **every ticket**, regardless of risk level, has a permanent record of:

- AI predictions
- Final resolution
- Email sent
- Queue assignment

3. Add a “Low Confidence” Path

If the AI confidence score is below a threshold (e.g., < 0.60), route the ticket to a dedicated path:

- **Low Confidence Path**
 - No automated email
 - Human review required
 - Additional notes captured

3. Add a “Low Confidence” Path

If the AI confidence score is below a threshold (e.g., < 0.60), route the ticket to a dedicated path:

- **Low Confidence Path**
 - No automated email
 - Human review required
 - Additional notes captured

4. Add a “Fraud / Security Alert” Path

If the AI detects keywords related to:

- fraud
- hacking
- unauthorized access
- suspicious activity

Create a dedicated path:

- **Fraud/Security Escalation Path**
 - Immediate internal alert
 - No automated customer email
 - Mandatory human review

5. Add SLA Tracking

Add timestamps and calculate:

- Time to triage
- Time to approval
- Time to resolution

6. Add Auto-Assignment to Agents

Based on category or priority, the system could automatically assign tickets to:

- Billing team
- Technical team
- Retention team
- High-risk specialists

7. Add a Dashboard for Analytics

Using Google Looker Studio or Excel, to visualize:

- Ticket volume by category
- Rejection vs approval rates
- High-risk trends
- Billing escalation frequency
- AI confidence distribution